STAT 447 Project 1 - Initial Draft

Alex White and Shelby Hennlich

2026-03-08

I. Introduction to Data

Parkinson's Disease, according the National Institute of Neurological Disorders and Stroke (NIH), is best classified as a disorder that attacks the nervous system, making it increasing difficult for an individual to perform movement-based tasks. In particular, Parkinson's Disease can have a severe effect on an individual's talking capability, often causing hesitation, slurred words, and monotony whilst speaking (NIH).

However, while it is known that Parkinson's Disease (PD) causes these issues in speaking, the specific medical differences in voice quality seen between those with PD and those without can also be investigated.

This leads to the primary research question that will be explored in this report:

Which voice measurements can successfully be used to discriminate healthy people from those that do struggle with PD?

To answer this question, the vocal fundamental frequencies and tonal components of individuals with and without PD can be explored.

For this specific report, the data of the vocal fundamental frequencies and tonal components was collected from 32 individuals, 24 of which were diagnosed with PD. Each individual provided around 6 vocal recordings, with a total of 195 voice recordings and their measurements being collected from the 32 individuals. Alongside this measurement-focused data, the demographics and stages of PD for the 24 diagnosed individuals and 8 healthy individuals was also collected.

With the purpose and the background of this PD data established, the exploration process of the 31 individuals and their distinguishing voice measurements can begin.

II. Exploring the Data

Before this data exploration, as each of the 32 individuals had multiple recordings, the data first had to be condensed from 195 observations to a representative 32. More specifically, the median value of each individuals' vocal measurements for their repeated recorded tests was chosen to protect the vocal measurements against outliers and skewness. After these median values were calculated, it was then necessary to join the patients' data with their demographic information.

With the condensed and joined data set created, it was then possible to explore relationships in the data using various EDA techniques:

Technique #1: Data Summaries

To begin, various medical and demographic qualities of the 32 patients were analyzed in the form of frequency tables:

Table 1: Frequency of Parkison's Disease(PD) Across Patients

PD Diagnosis	Count	Proportion (%)
Has PD	8	25
No PD	24	75

Table 2: Frequency of Females/Males Across Patients

Sex	Count	Proportion (%)
Female	12	38.71
Male	19	61.29

Table 3: Frequency of Ages Across Patients

Age Range	Count	Proportion (%)
40-49	3	9.68
50-59	2	6.45
60-69	14	45.16
70-79	10	32.26
80+	2	6.45

From these three frequency tables, the following observations were made:

- *Figure 1.1:* From the study, 75% of the patients did not have Parkinson's Disease at the time the data was collected.
- *Figure 1.2:* Of the 32 patients, 61.29% are male and only 38.71% are female (with one patient not having a listed sex).
- *Figure 1.3:* Each of the 32 patients are at least 40 years old, with 77.42% of patients being 60 to 79 years old (with one patient not having a listed age).

Technique #2: Variable Relationships

Next, the vocal measurements of the 32 patients were plotted against one another to see any inter-variable relationships without the impact of demographic and the Parkinson's diagnosis data:

From these four variable-relationship plots, the following observations were made:

- *Figure 2.1:* The plots for variables related to vocal fundamental frequency showed generally positive and slightly scattered relationships, indicating there is no unseen skewness or inconsistency in these variables from the data.
- *Figure 2.2:* The plots for variables related to variation in vocal fundamental frequency also showed generally positive but stronger relationships, indicating there is no unseen skewness or inconsistency in these variables from the data.
- *Figure 2.3:* The plots for variables related to variation in vocal amplitude showed similar results to those seen in *Figure 2.2,* indicating similar results of consistent data for these variables.

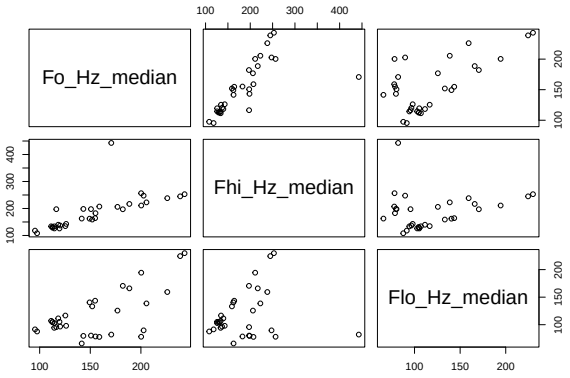

Figure 1: Figure 2.1: Variable Relationships of Vocal Fundamental Frequency

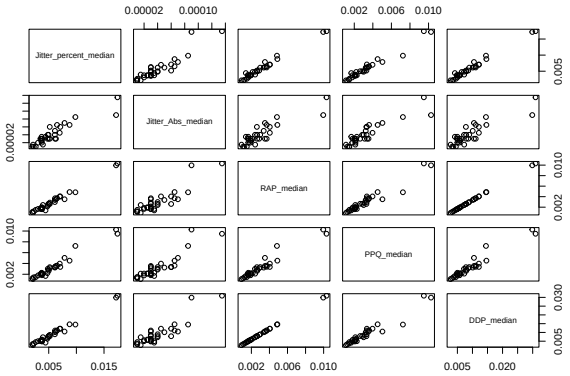

Figure 2: Figure 2.2: Variable Relationships of Variation in Vocal Fundamental Frequency

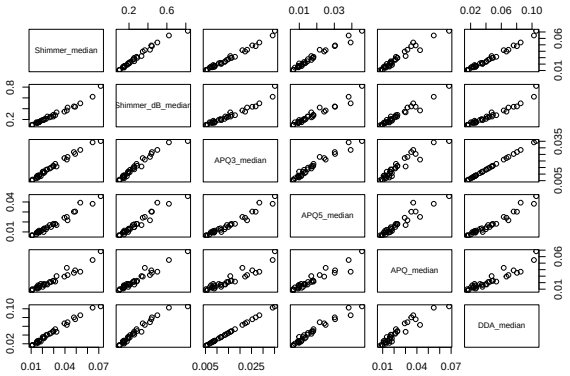

Figure 3: Figure 2.3: Variable Relationships of Variation in Vocal Amplitude

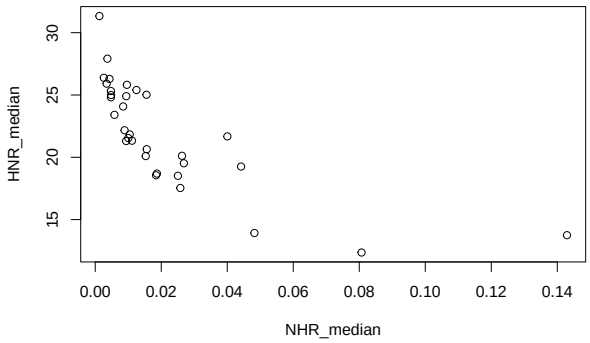

Figure 4: Figure 2.4: Variable Relationships of Ratio of Noise to Tonal Components in the Voice

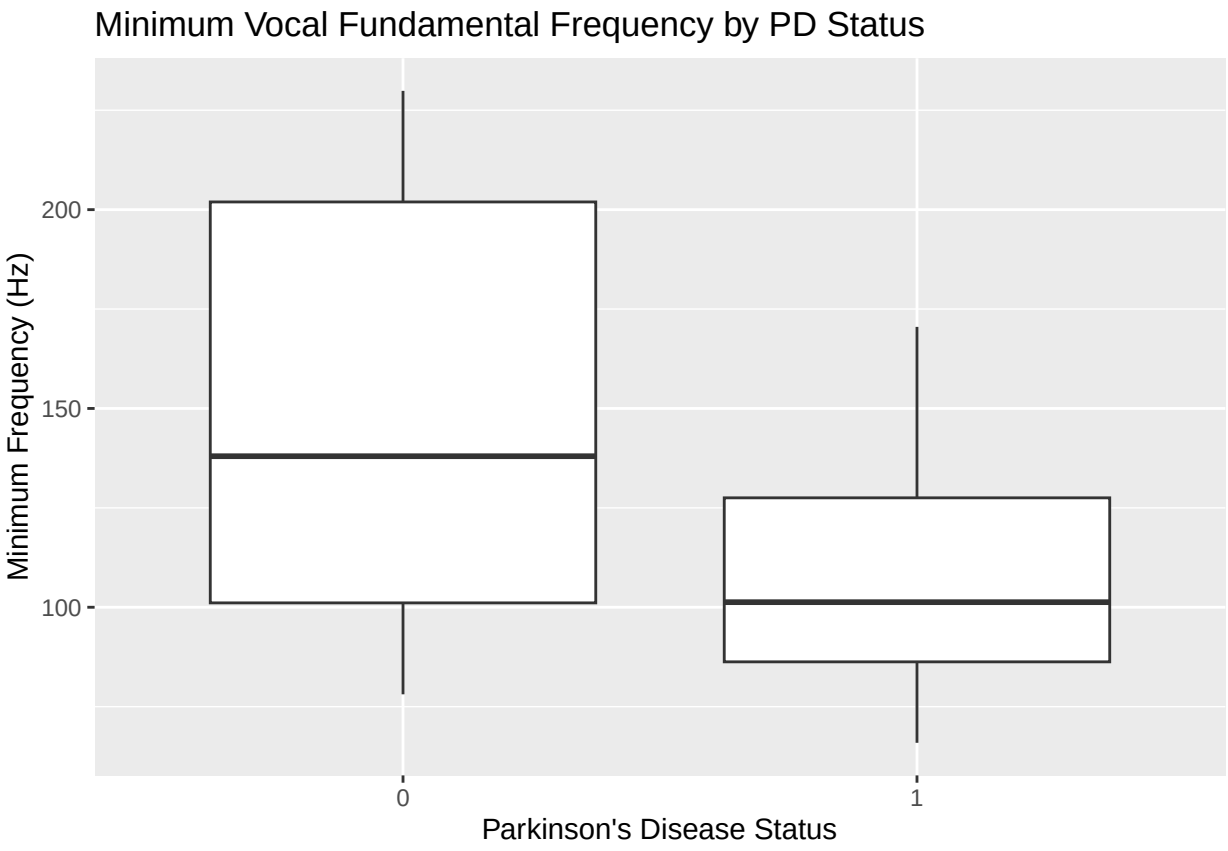

Figure 5: Figure 3.1: Boxplots of Vocal Measurements by Parkinson's Disease Status

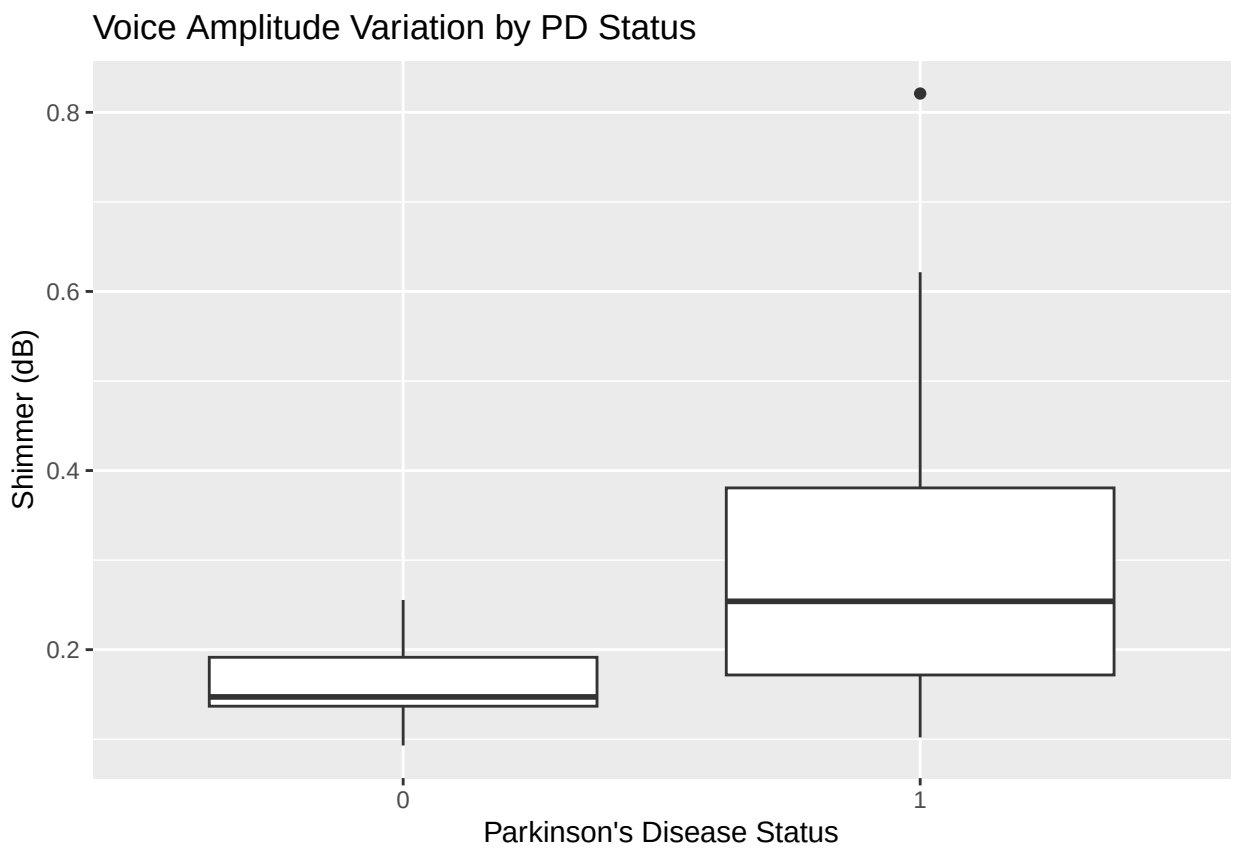

Figure 6: Figure 3.2: Voice Amplitude Variation by Parkinson's Disease Status

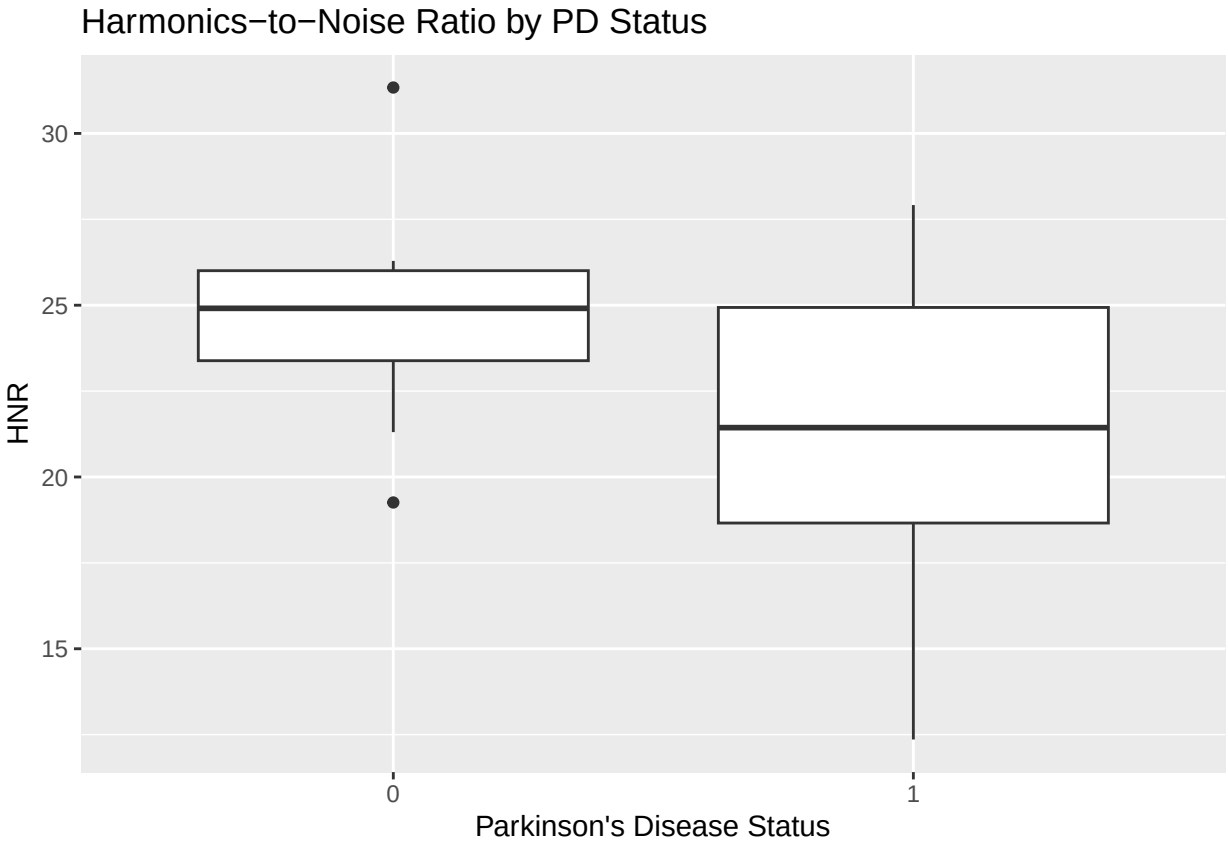

Figure 7: Figure 3.3: Harmonics-to-Noise Ratio by Parkinson's Disease Status

- *Figure 2.4:* The plot for the two variables measuring noise and tonal components in the voice showed a negative but strong relationship, but as the two variables are inverses of one another, remain consistent and not skewed.

Technique #3: Box Plots

Boxplots were created to compare key vocal measurements between individuals diagnosed with Parkinson's disease and healthy individuals. These visualizations allow for comparison of the median values, spread, and potential outliers between the two groups.

From the boxplots, individuals with Parkinson's disease appear to have lower minimum vocal frequencies, greater variation in voice amplitude, and lower harmonics-to-noise ratios compared to healthy individuals. These visual differences support the predictions that certain vocal characteristics may help discriminate between individuals with and without Parkinson's disease.

III. Statistical Methods

With the exploration of this data completed, our overall predictions of what voice measurements are the primary discriminators of those with or without PD were initialized.

Prediction #1: Individuals with a lower minimum vocal fundamental frequency (in Hertz) have a higher likelihood of having Parkinson's Disease.

Prediction #2: Individuals with higher voice amplitude (in decibels) have a higher likelihood of having Parkinson's Disease.

Prediction #3: Individuals with a lower HNR or Harmonics-to-Noise Ratio (in Hertz) have a higher likelihood of having Parkinson's Disease.

To test these three predictions, we utilized a nonparametric statistical approach for comparing voice measurements between individuals with PD and healthy individuals. Because we were comparing these measurements across only two populations (PD and non-PD), the primary statistical method used was the Mann-Whitney U Test.

IV. Results and Conclusions

```
##
##  Wilcoxon rank sum exact test
##
## data:  Flo_Hz_median by status
## W = 135, p-value = 0.09431
## alternative hypothesis: true location shift is not equal to 0
```

```
##
##  Wilcoxon rank sum exact test
##
## data:  Shimmer_dB_median by status
## W = 38, p-value = 0.01018
## alternative hypothesis: true location shift is not equal to 0
```

```
##
##  Wilcoxon rank sum exact test
##
## data:  HNR_median by status
## W = 138, p-value = 0.07036
## alternative hypothesis: true location shift is not equal to 0
```

Statistical Methods: Following the exploratory analysis, three predictions were developed regarding potential differences in voice measurements between individuals diagnosed with Parkinson's disease (PD) and healthy individuals. Specifically, we examined whether minimum vocal fundamental frequency, voice amplitude variation, and the harmonics-to-noise ratio differed between the two groups.

To test these predictions, a **nonparametric statistical approach** was used. Because the sample size was relatively small and the distribution of several voice measurements appeared skewed, the **Mann–Whitney U test** was used to compare measurements between individuals with PD and healthy individuals. This test evaluates whether the distributions of the measurements differ between the two groups and is appropriate for comparing medians between two independent samples.

A significance level of $\alpha = \mathbf{0.05}$ was used for all hypothesis tests.

Results and Conclusions:

Prediction 1: (Minimum Vocal Fundamental Frequency)

A Mann–Whitney U test was conducted to compare the minimum vocal fundamental frequency between individuals with Parkinson's disease and healthy individuals. The results indicate that there is not significant evidence that the median minimum vocal fundamental frequency differs between individuals with Parkinson's disease and healthy individuals ($W = 135$, $p = 0.094$) at the $\alpha = 0.05$ significance level.

Prediction 2: Voice Amplitude Variation

A Mann–Whitney U test was also conducted to compare voice amplitude variation between individuals with Parkinson's disease and healthy individuals. The results provide **significant evidence** that the median voice amplitude variation differs between individuals with Parkinson's disease and healthy individuals **($W = 38$, $p = 0.010$)** at the $\alpha = 0.05$ significance level.

This result suggests that variation in voice amplitude may be a useful characteristic for distinguishing individuals with Parkinson's disease from healthy individuals.

Prediction 3: Harmonics-to-Noise Ratio

A Mann–Whitney U test was conducted to compare the harmonics-to-noise ratio (HNR), a measure of voice clarity, between individuals with Parkinson's disease and healthy individuals. Although the p-value ($p = 0.070$) approaches significance, it does not meet the $\alpha = 0.05$ threshold. Therefore, we fail to reject the null hypothesis and conclude that there is insufficient evidence to suggest a difference in median HNR between the two groups ($W = 138$, $p = 0.070$).

Overall, the analysis explored whether several voice measurements could be used to distinguish individuals with Parkinson's disease from healthy individuals. Among the three vocal characteristics examined, only variation in voice amplitude showed a statistically significant difference between the two groups. Specifically, the Mann–Whitney U test indicated significant evidence that the median voice amplitude variation differs between individuals with Parkinson's disease and healthy individuals.

Although minimum vocal fundamental frequency and the harmonics-to-noise ratio showed some differences between the groups, these differences were not statistically significant at the $\alpha = 0.05$ level. These results suggest that certain vocal measurements, particularly amplitude variation, may serve as useful indicators for distinguishing individuals with Parkinson's disease from healthy individuals.

Future research could explore additional voice measurements or apply predictive modeling techniques to determine which combination of vocal features provides the most accurate classification of Parkinson's disease.